1 Inline and display mathematics

A sentence with inline mathematics: $y = mx + c$.

The same formula in display mode:

$$y = mx + c$$

Superscripts and subscripts: a^b, a_b, x^2, x_i.

Greek letters and mathematical functions: α, β, θ, Γ, Δ, $y = 2\sin\theta^2$, $\log x$.

2 Numbered equation

An integral with an automatically generated number:

$$\int_{-\infty}^{+\infty} e^{-x^2}\,dx \tag{1}$$

3 Aligned equations

A recurrence without equation numbers:

$$Q_{n,0} = 1 \quad Q_{0,k} = [k = 0];$$
$$Q_{n,k} = Q_{n-1,k} + Q_{n-1,k-1} + \binom{n}{k}, \quad \text{for } n,\, k > 0.$$

Several columns of aligned equations:

$$\begin{aligned} a &= b + 1 & c &= d + 2 & e &= f + 3 \\ r &= s^2 & t &= u^3 & v &= w^4 \end{aligned}$$

4 Matrices

Three matrix styles:

$$\begin{matrix} a & b & c \\ d & e & f \end{matrix} \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \quad \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

Right-aligned matrix columns using mathtools:

$$\begin{pmatrix} 10 & 11 \\ 1 & 2 \\ -5 & -6 \end{pmatrix}$$

5 Other multiline environments

Centered equations without alignment:

$$P(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + f \tag{2}$$

$$x^2 + x = 10 \tag{3}$$

A single expression split across several lines:

$$\begin{aligned}(a+b+c+d)x^5 + (b+c+d+e)x^4 \\ + (c+d+e+f)x^3 + (d+e+f+a)x^2 + (e+f+a+b)x \\ + (f+a+b+c)\end{aligned}$$

An aligned block inside a larger expression:

$$\begin{cases} a = b \\ c = d \end{cases}$$

6 Mathematical fonts

ABC ABC $\mathbf{ABC}$ ABC $\mathtt{ABC}$ $\mathbb{ABC}$

Comparing several ways of writing a word:

$size \neq \mathit{size} \neq \mathrm{size}$

Nested font commands:

ABC $\mathbf{ABC}$

7 Bold mathematics

Normal weight: $(x+y)(x-y) = x^2 - y^2$.

A bold expression: $\boldsymbol{(x+y)(x-y) = x^2 - y^2}$.

Individual bold Latin letters: $(x+\mathbf{y})(x-\mathbf{y}) = x^2 - \mathbf{y}^2$.

Bold Greek letters using bm: $\alpha + \boldsymbol{\alpha} < \beta + \boldsymbol{\beta}$.